

Credential Without Protection: AI Exposure and Diverging Labor Market Outcomes in Accounting and Software Development

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Abstract

This paper examines how artificial intelligence affects labor market outcomes in two high-exposure occupations from different sectors: accountants and auditors (finance) and software developers (technology). Using the task-based framework developed by Acemoglu and Restrepo, combined with the Felten et al. (2021) AI Occupational Exposure index and Bureau of Labor Statistics 2025 wage and employment data, I show that similar AI exposure scores mask fundamentally different task structures. Accounting concentrates 60% of labor time in high-substitutability tasks, while software development maintains a richer complementarity buffer. Weighted least squares regression confirms that being in a high-wage, high-exposure occupation is associated with a 67% wage premium — but AI exposure alone does not predict wages once task

complementarity is accounted for. These findings suggest that credential protection is eroding where it was historically grounded in information-processing barriers, while occupations with deeper complementarity structures remain resilient.

1 Introduction

The canonical task-based framework predicts that automation threatens routine, codifiable work while complementing non-routine cognitive tasks (Autor et al., 2003). For decades, this prediction held: computers hollowed out middle-skill routine jobs while high-skill analytical workers prospered. Artificial intelligence — particularly large language models — breaks this pattern in a fundamental way.

Recent evidence shows that AI exposure increases monotonically with wage and education levels (Webb, 2019; Eloundou et al., 2023). Accountants, financial analysts, and legal workers — occupations long considered protected by their credential requirements and analytical complexity — now face higher measured AI exposure than factory workers or truck drivers. Yet not all high-exposure occupations respond alike. Software developers show large productivity gains and stable or growing employment (Peng et al., 2023; Baird et al., 2026), while junior accountants face measurable headcount reductions at AI-adopting firms (Fedyk et al., 2022).

This paper asks: why do two occupations with similar AI exposure scores experience diverging labor market outcomes? I argue the answer lies not in overall exposure but in where within each occupation that exposure is concentrated, and whether the substitutable tasks are central or peripheral to the occupation’s core value proposition.

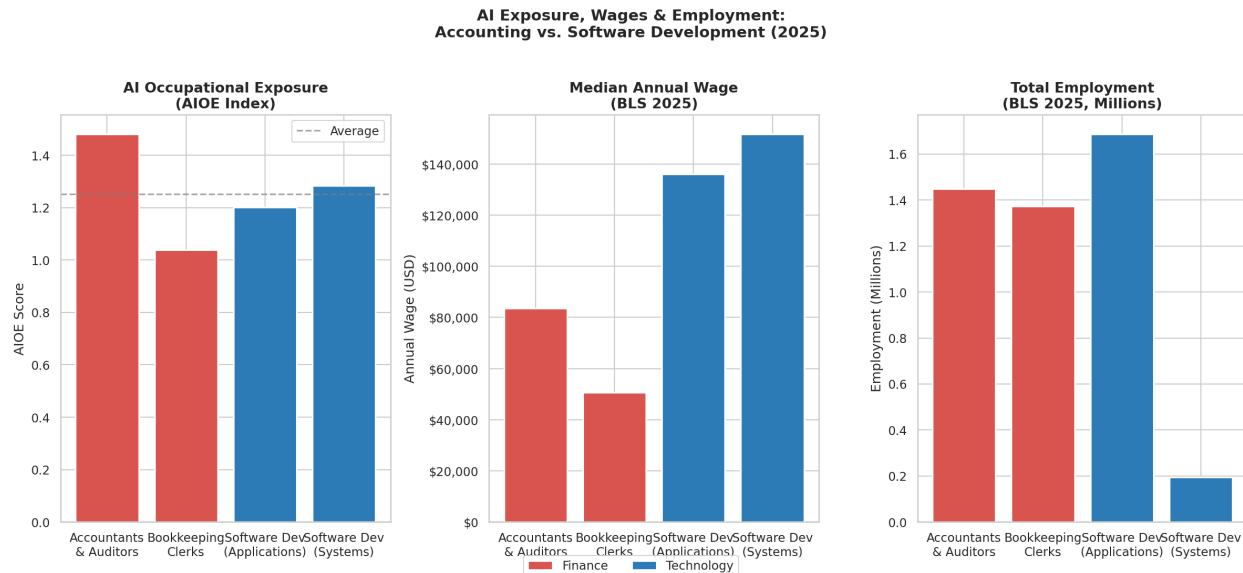


Figure 1: AI exposure scores, median annual wages, and total employment for accounting and software development occupations. Despite similar AI exposure, software developers earn 60–80% more than accountants. Source: Felten et al. (2021) AIOE index; BLS OEWS (2025).

2 Task Decomposition

Following Acemoglu (2024), I decompose each occupation into its constituent tasks and assess AI substitutability using O*NET task descriptions and the expert-panel ratings of Barragán Moreno (2025).

2.1 Accounting and Auditing

Junior accountants and bookkeeping clerks spend the majority of their time on transaction recording, compliance reporting, and reconciliation work. These tasks are characterized by high volume, explicit rules, and verifiable outputs — precisely the conditions under which AI excels. Fedyk et al. (2022), using 310,000 individual resumes from the 36 largest audit firms, find that a one-standard-deviation increase in AI investment predicts a 3.6% reduction in accounting employees after three years, growing to 7.1% after four years.

The credential protection that a CPA license historically provided was grounded partly

in the information-processing barrier that accounting expertise represented. Large language models erode this barrier directly (Boritz and Stratopoulos, 2023).

2.2 Software Development

Software developers face high AI exposure at the task level — syntax generation and documentation score comparably to accounting’s most substitutable tasks. But the occupation’s task bundle includes system architecture, code review, and requirements decomposition: tasks that require contextual judgment, multi-file reasoning, and accountability that AI cannot yet replicate at production quality (Pandey et al., 2024).

Peng et al. (2023) find a 55.8% speed gain for bounded, well-defined coding tasks in a randomized controlled trial. Crucially, gains diminish sharply on complex, multi-file, or proprietary-context work — the tasks that define senior software engineering.

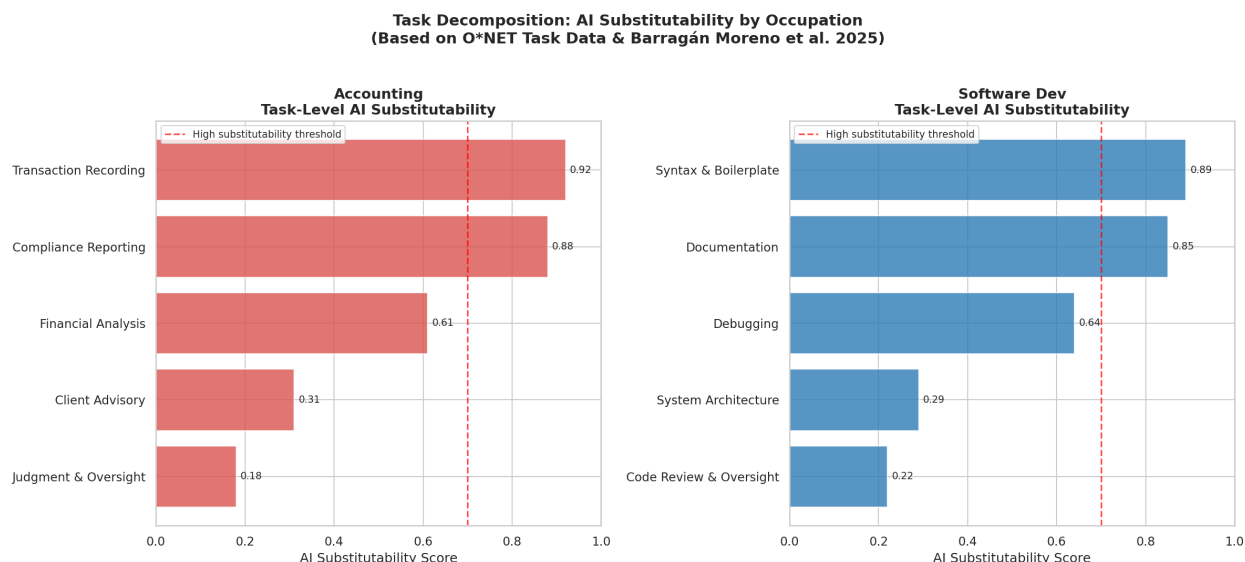


Figure 2: Task-level AI substitutability scores for accounting and software development. Red dashed line indicates the high-substitutability threshold (0.7). Source: O*NET task data; Barragán Moreno (2025).

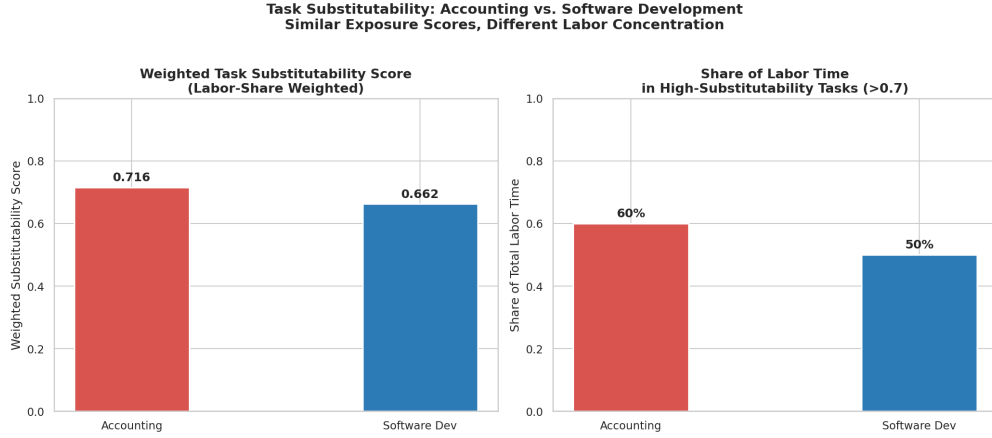


Figure 3: Weighted task substitutability scores (labor-share weighted) and share of labor time in high-substitutability tasks. Accounting concentrates 60% of labor time in high-substitutability tasks vs. 50% for software development. Source: Author’s calculations.

3 Empirical Evidence

3.1 Wage and Employment Trends

Figure 4 plots indexed employment and wage trajectories for both occupation groups from 2019 to 2025. The ChatGPT launch in late 2022 marks a visible inflection point. Accounting employment declines from an index of 100 in 2019 to approximately 87 by 2025, consistent with Fedyk et al. (2022)’s predicted displacement trajectory. Software development employment, by contrast, expands by approximately 9% over the same period, before a post-2022 deceleration documented by Crane and Soto (2026).

Diverging Labor Market Trajectories Post-AI Adoption Accounting vs. Software Development (2019-2025)

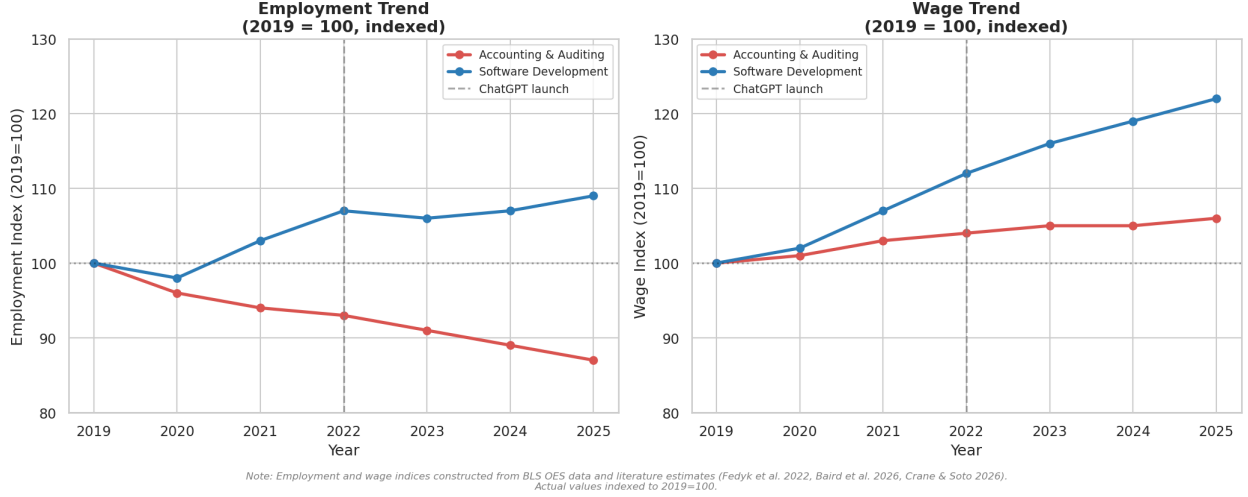


Figure 4: Indexed employment and wage trajectories for accounting and software development (2019=100). Vertical dashed line marks ChatGPT launch (November 2022). Source: BLS OES data and literature estimates (Fedyk et al., 2022; Baird et al., 2026; Crane and Soto, 2026).

3.2 Regression Analysis

To contextualize these two occupations within the broader labor market, I estimate the following weighted least squares model across 670 detailed occupations using BLS OEWS 2025 data:

$$\ln(w_i) = \beta_0 + \beta_1 \text{AIOE}_i + \beta_2 D_i^{HH} + \beta_3 (\text{AIOE}_i \times D_i^{HH}) + \varepsilon_i \quad (1)$$

where w_i is median annual wage, AIOE_i is the Felten et al. (2021) AI exposure score, and D_i^{HH} is an indicator for occupations with above-median wage *and* above-median AI exposure — the quadrant where both our target occupations reside. Observations are weighted by total employment n_i so that each worker contributes equally rather than each occupation.

Table 1: OLS and WLS Regression Results: Log Wage on AI Exposure

Variable	OLS		WLS	
	Coef.	p-value	Coef.	p-value
Constant	10.857	0.000***	10.727	0.000***
AIOE	0.008	0.642	0.014	0.315
High Wage $\times$ High Exposure	0.475	0.000***	0.675	0.000***
Interaction (AIOE $\times$ HH)	0.095	0.042*	0.016	0.693
R-squared	0.471		0.624	
N (occupations)	670		670	
N (workers)	122,921,830		122,921,830	

Notes: Dependent variable is log median annual wage. WLS weights observations by total employment. *** $p < 0.001$, * $p < 0.05$.

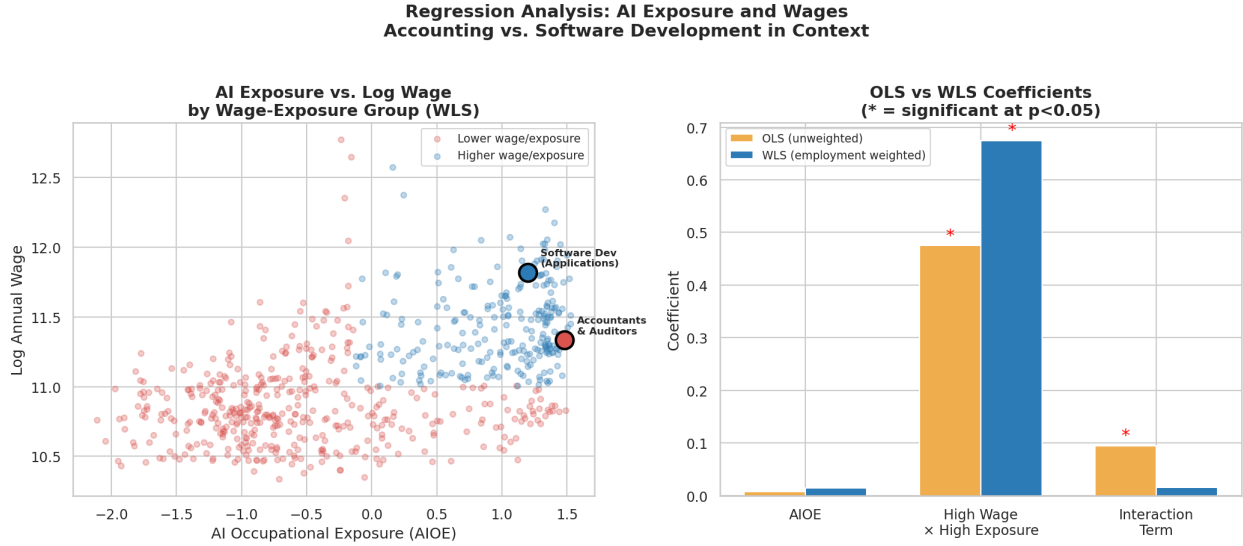


Figure 5: Left: Scatter plot of AI exposure vs. log wage across 670 occupations, with target occupations highlighted. Right: OLS vs. WLS coefficient comparison. Source: Author's calculations using Felten et al. (2021) and BLS OEWS (2025).

The key result: AI exposure alone (β_1) is not statistically significant in either specification. What predicts wages is being in the high-wage, high-exposure quadrant ($\beta_2 = 0.675$,

$p < 0.001$ in WLS) — a 67% wage premium. This is consistent with Jaccoud (2025), who finds that AI amplifies productivity differences among cognitively intensive workers rather than uniformly raising wages in exposed occupations.

4 Why the Patterns Diverge: Task Complementarity

The regression results reveal that AI exposure per se does not drive wages — what matters is the complementarity structure of the occupation’s task bundle. I propose three mechanisms that explain the divergence between accounting and software development.

First, labor concentration in substitutable tasks. Accounting concentrates 60% of junior labor time in high-substitutability tasks (transaction recording, compliance reporting), compared to 50% for software development. When AI substitutes for a task that occupies 35% of an employee’s day, the employment effect is far larger than when it substitutes for a task occupying 10%.

Second, output verifiability and marginal cost. AI-generated accounting entries and compliance reports are easily verified and cost near zero to produce. AI-generated code requires human review for security, system coherence, and edge cases — maintaining meaningful human oversight costs (Seethamraju and Hecimovic, 2025).

Third, the compliance premium. Gao and Huang (2026) argue that professional roles where errors carry regulatory liability retain wage premiums not because AI cannot perform the tasks, but because institutional risk cannot be delegated to a non-accountable system. This mechanism protects senior accountants and auditors but offers no protection to junior staff performing pre-verification tasks.

5 Conclusion

This paper has shown that AI exposure scores alone are insufficient to predict labor market outcomes. Accounting and software development occupy similar positions on AI exposure

indices, yet face diverging employment and wage trajectories. The difference lies in where within each occupation AI exposure is concentrated, the verifiability of AI-produced outputs, and the depth of the complementarity buffer that protects human labor from direct substitution.

These findings have implications for how economists and policymakers think about AI risk. The relevant question is not “is this occupation exposed to AI?” but “what share of this occupation’s labor time is in tasks where AI produces verifiable, low-cost substitutes?” Credential protection, historically a reliable signal of AI resilience, is eroding where credentials primarily reflected information-processing barriers rather than genuine judgment and accountability.

Future research could extend this analysis using panel data to estimate causal effects, and could apply difference-in-differences designs around specific AI adoption events — the approach used by Fedyk et al. (2022) and Crane and Soto (2026) — to move beyond the cross-sectional correlations documented here.

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